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THE SINGLE ENTITY ACCESS DESIGN VIEW

ISE305 – DATABASE DESIGN & PRACTICE

SOME SLIDE MATERIAL FROM WATSON

Session Objectives

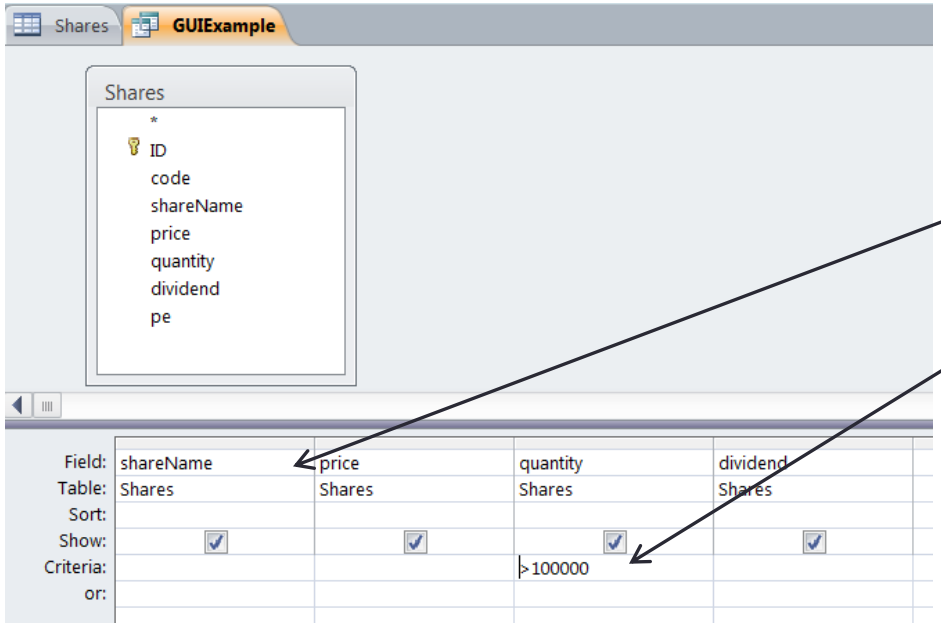
- Become familiar with the MS Access Design View for the development of queries

Access Design View

- Some DBMS provide a GUI query builder
- GUI query builders are usually
  - Not as powerful as pure SQL
  - Generate poorer performing SQL
  - Easier to code than SQL
- In Access, the query builder is referred to as the Design View

Access Design View Example

```
SELECT shareName, price, quantity, dividend
    FROM Shares WHERE quantity > 100000;
```



Chose the table, then the columns to display, then the criteria

shareName	price	quantity	dividend
Burmese Elephant	$0.07	154,713.00	$0.01
Bolivian Sheep	$12.75	231,678.00	$1.78
Royal Ostrich Farms	$33.75	1,234,923.00	$3.00

Are We on Track?

- List the names and prices of all shares with a price of at least $10.

shareName	price
Freedonia Copper	$27.50
Patagonian Tea	$55.25
Abyssinian Ruby	$31.82
Sri Lankan Gold	$50.37
Indian Lead & Zinc	$37.75
Bolivian Sheep	$12.75
Nigerian Geese	$35.00
Canadian Sugar	$52.78
Royal Ostrich Farms	$33.75

Were We on Track?

- Using Access Design View, list the names and prices of all shares with a price of at least $10.

Field:	shareName	price
Table:	Shares	Shares
Sort:		
Show:	✔	✔
Criteria:		>=10
or:		

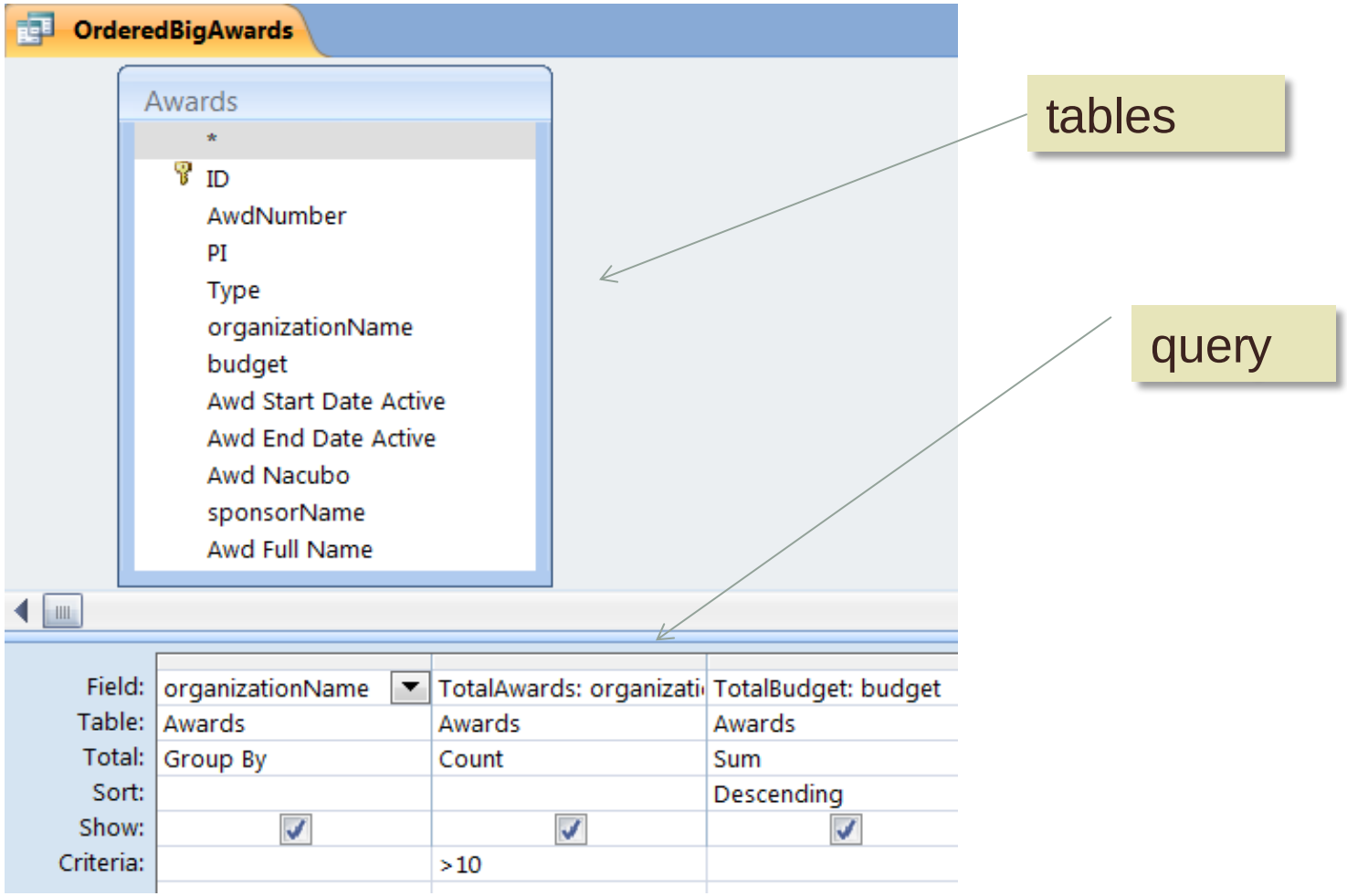
Were We on Track?

- SQL (generated by Access)

```
SELECT Shares.shareName, Shares.price
FROM Shares
WHERE (((Shares.price)>=10));
```

Notice the redundant SQL

GUI Interface Components



Design View - Sorting

- Drop-down allows you to select ascending or descending

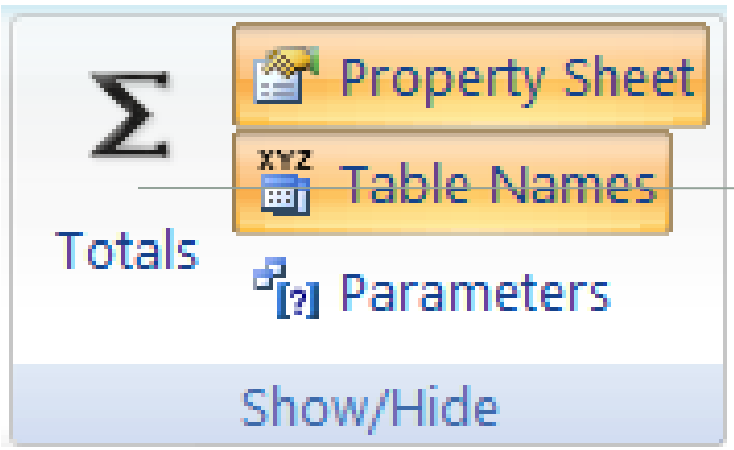
Field:	shareName	pe
Table:	Shares	Shares
Sort:		Ascending
Show:	✓	✓
Criteria:		

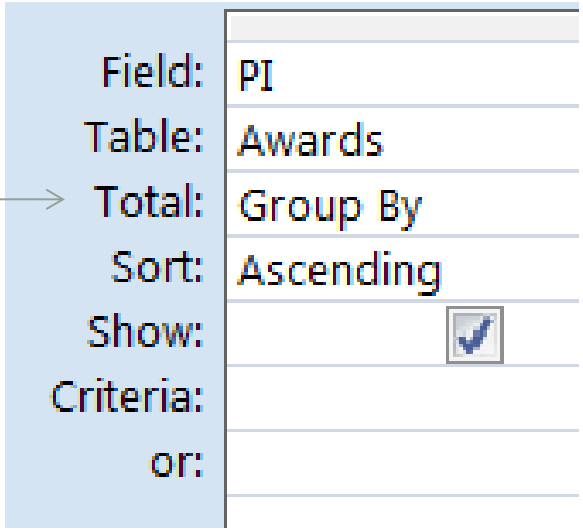
Shares | SortExample

shareName	pe
Burmese Elephant	3
Royal Ostrich Farms	6
Nigerian Geese	10
Patagonian Tea	10
Bolivian Sheep	11
Indian Lead & Zinc	12
Abyssinian Ruby	13
Canadian Sugar	15
Sri Lankan Gold	16
Freedonia Copper	16

Totals Selection

- In the Access Design menu, click on Totals to include totals in the query items





Access Design View

- Notice that when you click on Totals, the default in that row of the design view is GROUP BY
- You can select a function in the cell drop-down

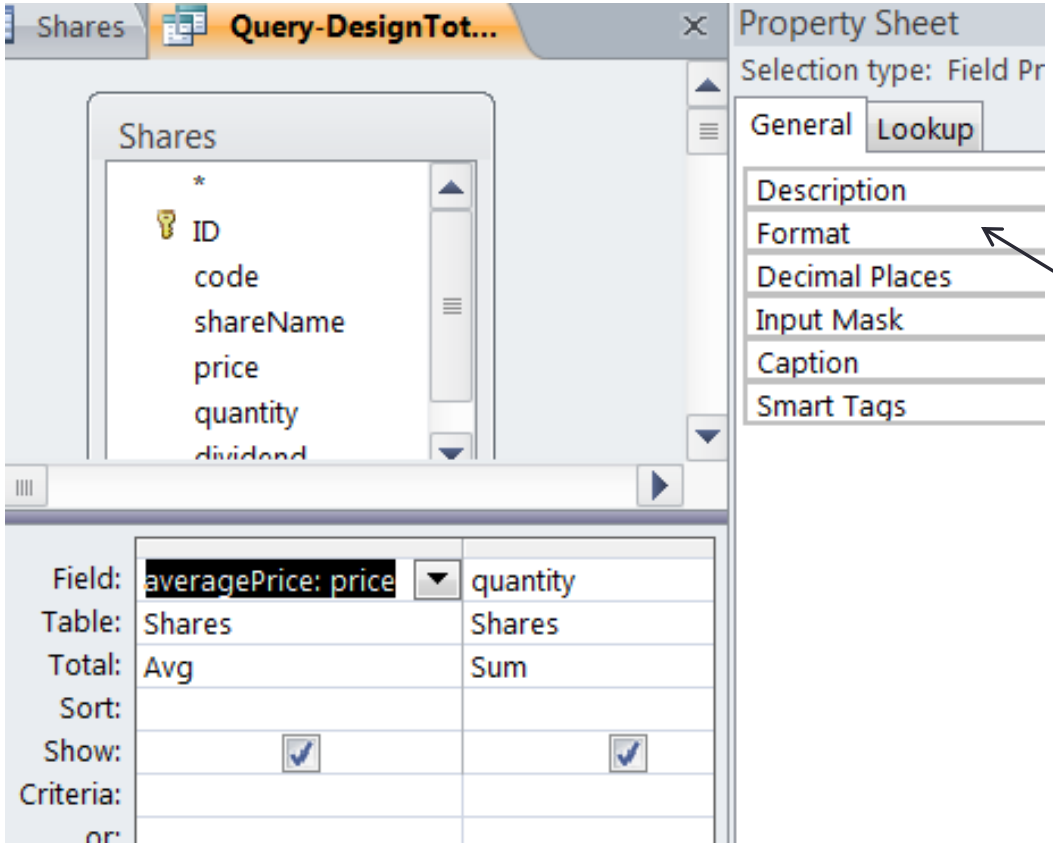
Field:	total: shareName	averagePrice: price	totalShares: quantity
Table:	Shares	Shares	Shares
Total:	Count	Avg	Sum
Sort:			
Show:	✓	✓	✓

total	averagePrice	totalShares
10	$33.70	1,722,785

You can also change the column header and format in Design View

Changing the Result Appearance

- Change the format of a column by using the Design View Property Sheet



Set Format to Currency

Are We on Track?

Remember to do this in steps

- Using the SBU Awards DB and Access Design View
  - Display the total budget of National Science Foundation awards by department
  - Order the result by budget

orgName	TotalBudget
050 Interschool Unit Programs	$23,961,482
050 Computer Science	$13,279,890
050 Chemistry	$7,332,631
050 Office of the Dean SOMAS	$6,360,256
050 Marine Sci Research Center	$5,453,212
050 Physics	$4,979,972
050 Interdepartmental - CEAS	$4,601,270
050 Mathematics	$4,581,299
050 Interdepartmental - Arts & Sciences	$3,671,626

HAVING Clause in Design View

- Aggregate function used

Restricted to groups with more than 60 awards

Field:	orgName	totalAwards: awdNumber
Table:	Awards	Awards
Total:	Group By	Count
Sort:		Descending
Show:	✓	✓
Criteria:		>60
or:		

orgName	totalAwards
050 Interschool Unit Programs	107
050 Chemistry	83
050 Physics	74
050 Pediatrics	64
050 Computer Science	63

```
SELECT Awards.orgName,
Count(Awards.awdNumber) AS
totalAwards
FROM Awards
GROUP BY Awards.orgName
HAVING
(Count(Awards.awdNumber))>60
ORDER BY Count(Awards.awdNumber)
DESC;
```

Are We on Track?

- Modify previous Track to:
  - Include the count of the number of awards
  - Only display organizations with more than 10 awards

orgName	numAwards	TotalBudget
050 Interschool Unit Programs	24	$23,961,482
050 Computer Science	35	$13,279,890
050 Chemistry	20	$7,332,631
050 Office of the Dean SOMAS	18	$6,360,256
050 Marine Sci Research Center	11	$5,453,212
050 Physics	17	$4,979,972
050 Mathematics	27	$4,581,299

Did You Meet Session Objectives?

- Become familiar with the MS Access Design View for the development of queries